Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			A chloroplast B nucleus NOT nucleolus	1			1		1
	(b)	(i)		Any **four** (x1) from: A. (The cytoplasm of) the amoeba has a {lower/ more negative} water potential than the {surrounding water / external environment}/ ORA (1) B. Water will therefore move into the cell by osmosis. (1) If not credited here, can be given with reference to vacuole C. (By pumping ions into the vacuole) {it lowers the water potential of the vacuole / creates a water potential gradient between the cytoplasm and the vacuole} (1) D. Water will move from the {cytoplasm/ cell} to the vacuole (by osmosis) (1) E. (The water is then expelled from the cell) which prevents {osmotic lysis/ bursting} (1) Reject reference to turgidity		2	2	4		
		(ii)		Mitochondria (1) To provide the ATP for {active transport of ions / pumping of ions/ exocytosis}(1) Ribosomes(1) Produce proteins for {transport of ions/ carrier proteins}(1)		4		4		
	(c)			• (The cell wall is) {strong / resists tension/ generates pressure potential/ owtte} **and** prevents {osmotic lysis/ bursting} (1) • Beta glucose (1) • It is made of (long) straight chains / alternate molecules {inverted/ rotated 180^{o} }(1) • {hydrogen bonds / cross-links} between (parallel) chains/ to form microfibrils (1)	2	2		4		

<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks Available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td>(d)</td><td></td><td></td><td>
<table>
<tr><th>Chlorella</th><th>Cyanophyceae</th></tr>
<tr><td>Membrane bound organelle/ named organelle present</td><td>No membrane bound organelles / named organelle</td></tr>
<tr><td>Linear DNA/ plasmid absent</td><td>{Loop of/ circular} DNA/ plasmid present</td></tr>
<tr><td>Larger/80s ribosomes</td><td>Smaller/70s ribosomes</td></tr>
<tr><td>Cellulose cell wall</td><td>Murein/peptidoglycan cell wall</td></tr>
<tr><td>Mesosome absent</td><td>Mesosome present</td></tr>
<tr><td>Flagellum absent</td><td>Flagellum present</td></tr>
<tr><td>Capsule absent</td><td>Capsule present</td></tr>
</table>
</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 2 total</td><td>5</td><td>8</td><td>2</td><td>15</td><td>0</td><td>1</td></tr>
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